



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

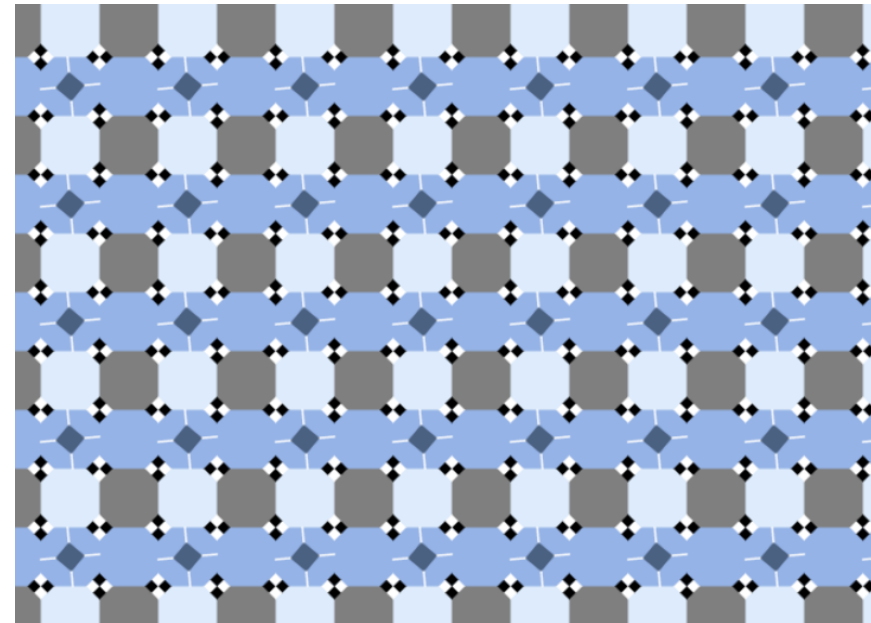
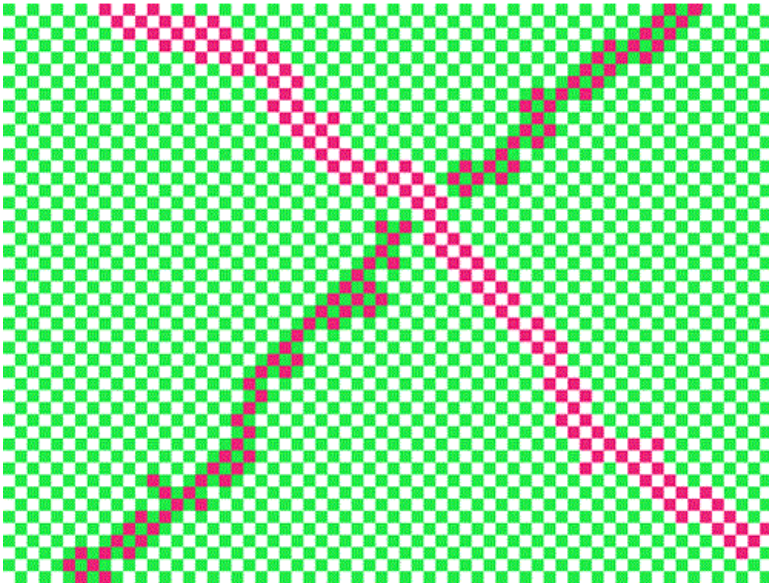
Lecture 17: Vision Processing, Cameras, Blob Detection

Marius Juston

DATE

Talk to the person next to you and answer these questions.

- What does RGB stand for?
- How many colors are there? Are the lines straight?



check out <https://michaelbach.de/ot/> for a bunch of optical illusions!

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

Lab: Starting Lab 6 this week. This a 3-week lab!

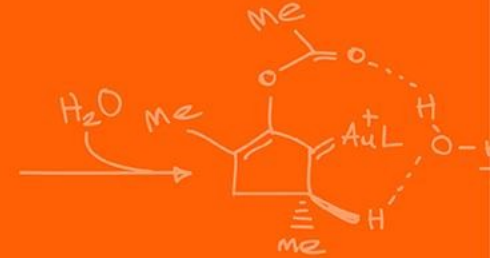
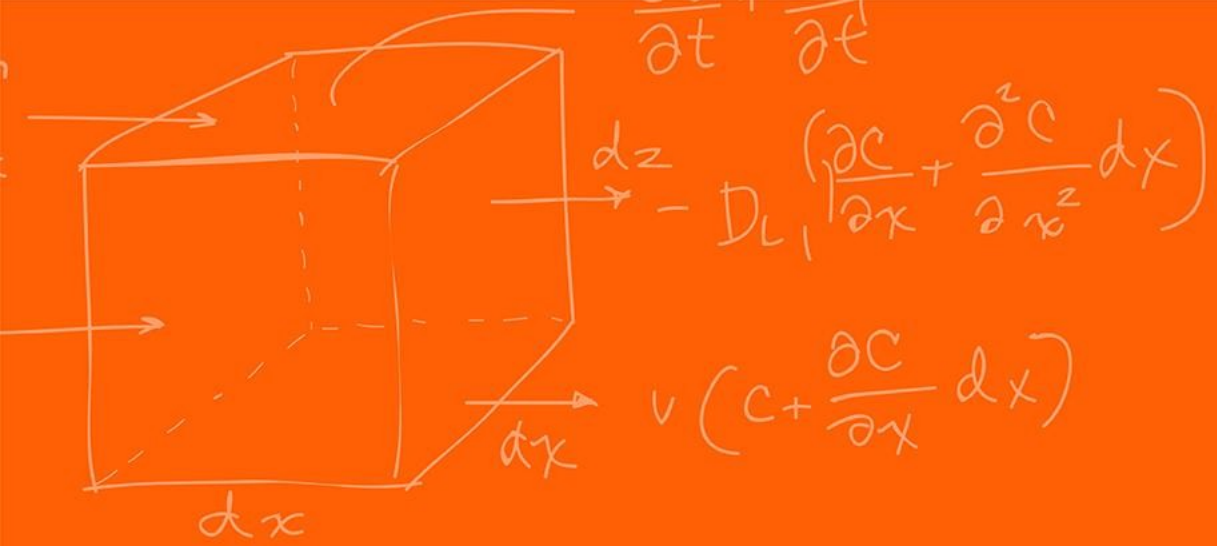
- Don't forget to have the Lab LabVIEW application implemented before lab.

Homeworks:

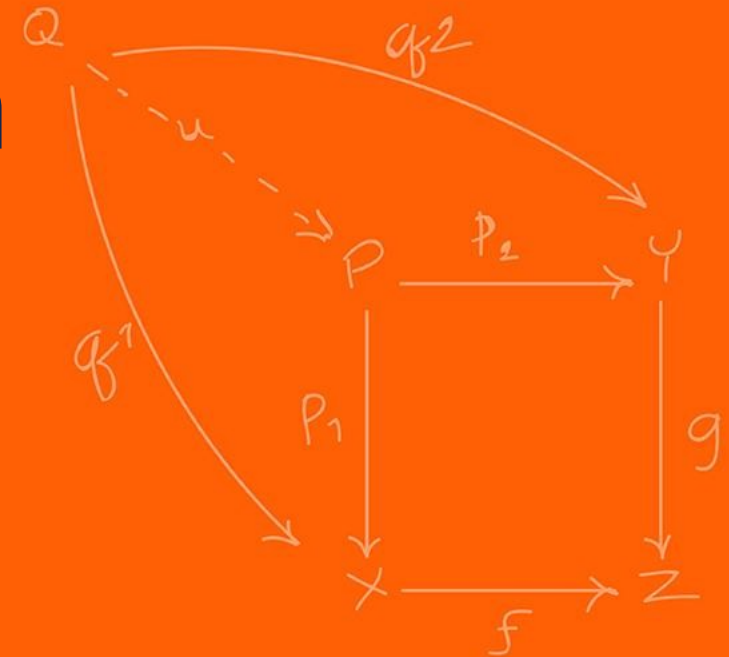
- Check-offs for [homework 4](#) Ex 2 and 3 are due **March 31, 5PM (The end of office hours)**
- Check-offs for [homework 4](#) Ex 4 are due **April 7, 5PM (The end of office hours)**
- Remainder of [homework 4](#) due on Gradescope on **April 8, 9AM**

Questions?

Homework, Lab, LabVIEW?

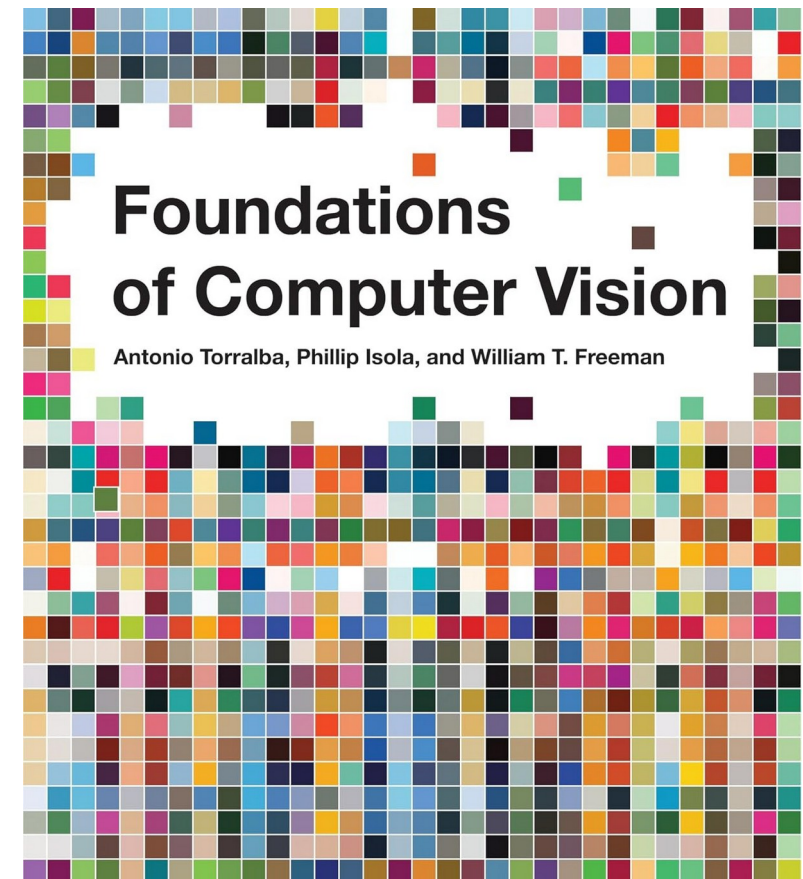


Computer Vision



I would **HIGHLY** recommend you looking at:

- <https://visionbook.mit.edu/>
- The webpage is a textbook with all the math, background and information for computer vision.
- We will only be doing a tiny portion of this in this lecture (there are whole grad classes for computer vision).
- It is containerized very nicely, has lots of pictures, diagrams. Each section takes 5 minutes to read.
- (also, for more CMOS details look at <https://isl.stanford.edu/~abbas/ee392b/lect04.pdf>)



Currently we have the LiDAR working and returning points; however, what kind of data are missing from just the LiDAR?

Color, complex shapes, human vision like data

Vision processing allows a robot to extract rich, high-density data from its environment.

Instead of an array of 228 distances (like our LiDAR), a camera gives us millions of data points per frame (RGB per pixel).

What is the primary drawback of having millions of data points per frame in a real-time mechatronic system?

Computational latency; processing takes time, more memory / bandwidth is required to handle processing of the data!

Is a standard camera an active or a passive sensor?

Passive.

It does not emit energy; it captures ambient light bouncing off objects.

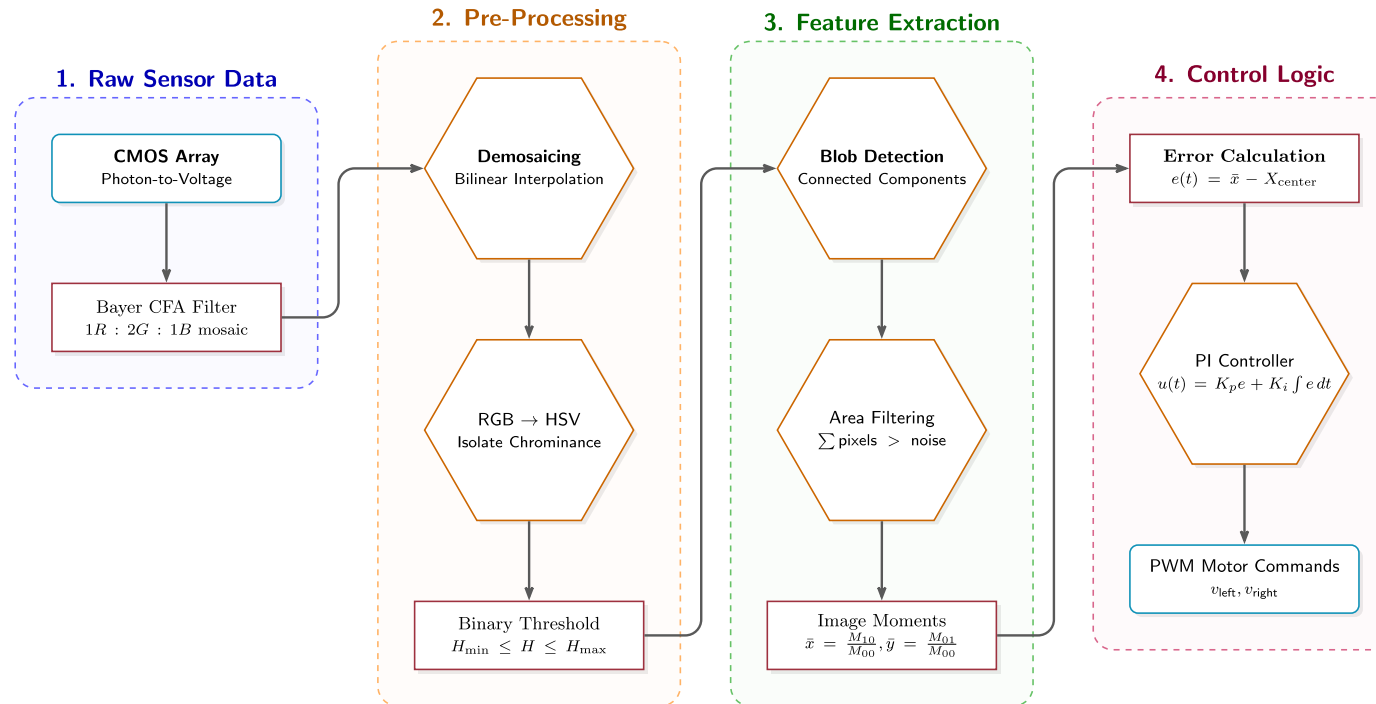
Because it is passive, it is highly vulnerable to environmental changes.

If the sun goes behind a cloud what happens?

The pixel color changes, shadows appears, etc.

How do we build robust code around this?

The main pipeline that we will follow in this class is the following:

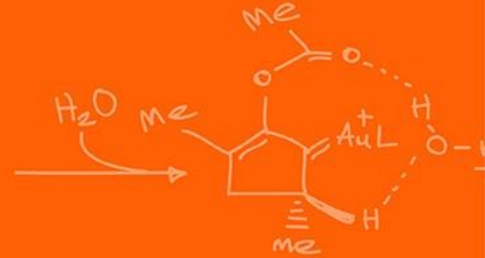
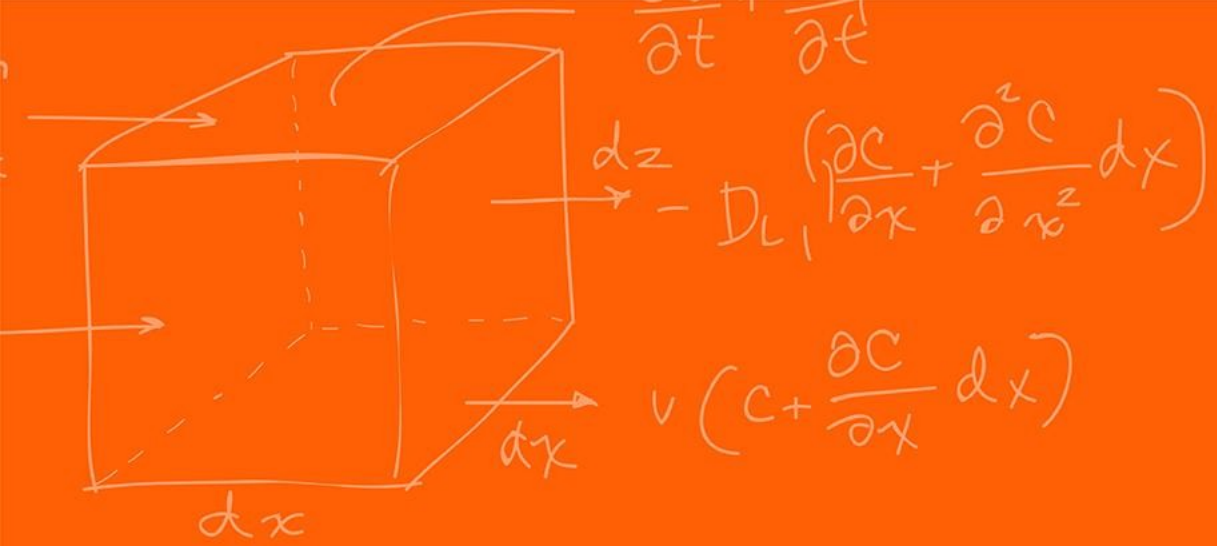


Step 1: The CMOS sensor captures photons and converts them to digital arrays.

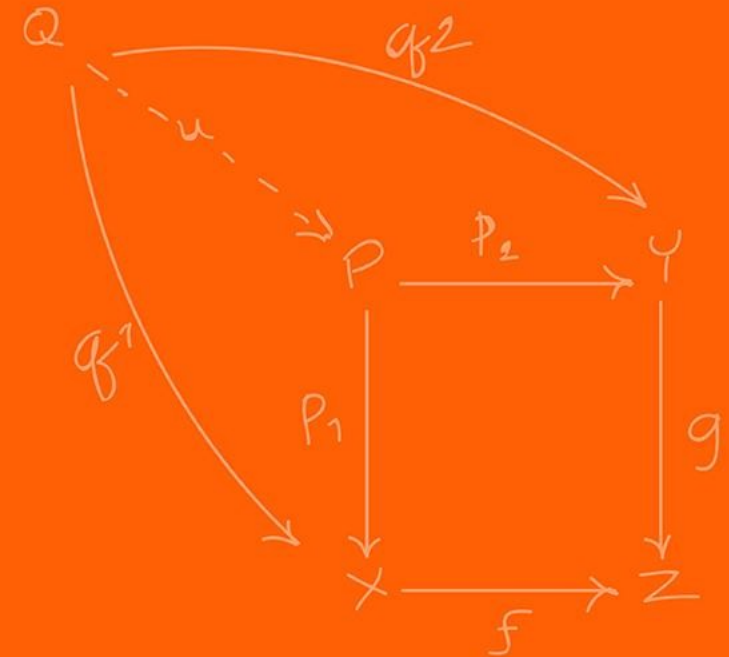
Step 2: The processor filters the image (e.g., finding specific colors).

Step 3: The algorithm calculates the coordinates of the target (Centroid).

Step 4: The Cascade Control loop turns the coordinates into wheel velocities.



CMOS



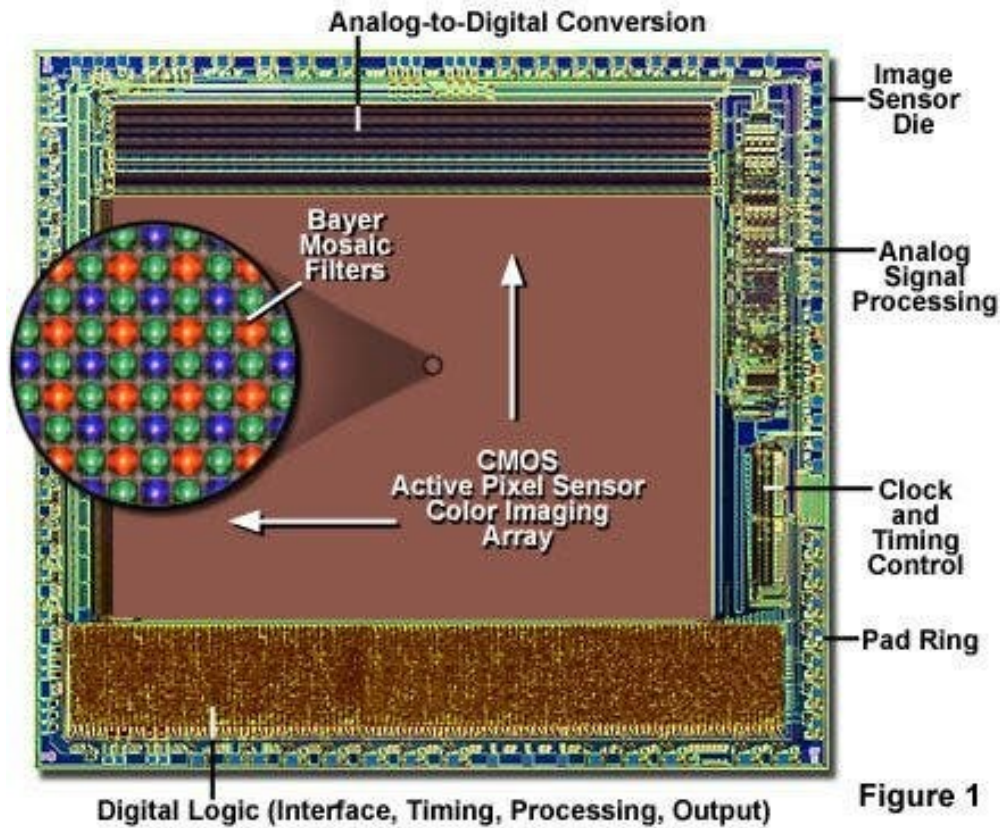
Most modern robotics vision relies on CMOS (Complementary Metal-Oxide-Semiconductor) sensors.

A CMOS sensor contains a massive 2D grid of microscopic photodiodes.

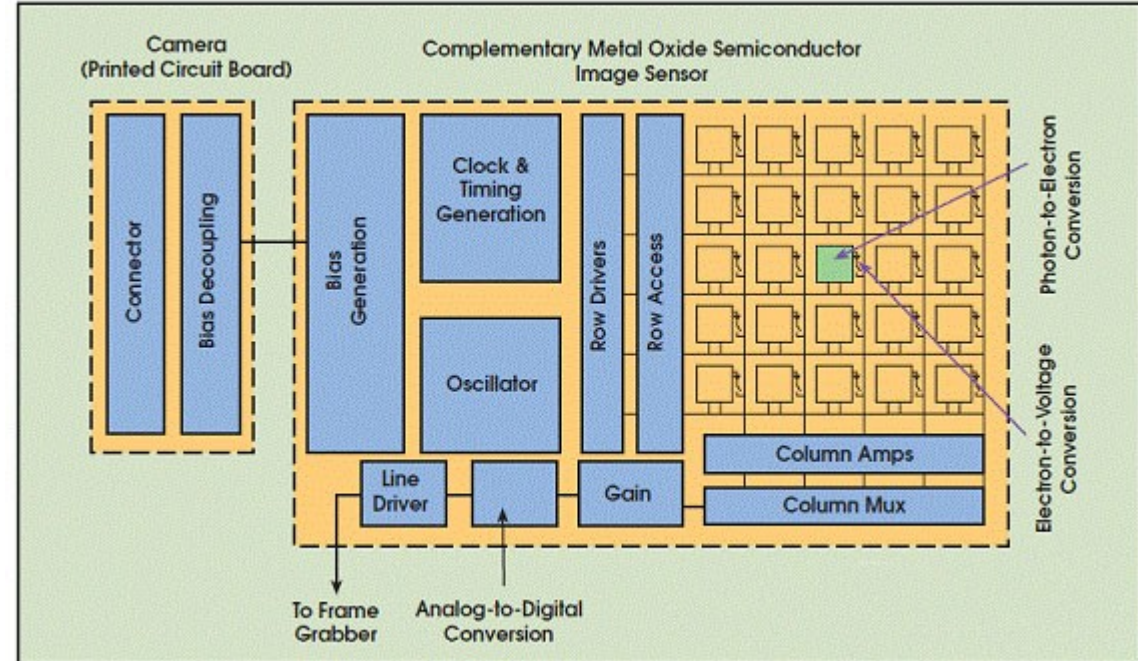
When photons strike a photodiode, they excite electrons via the photoelectric effect, creating a measurable voltage.

This continuous voltage is sampled and converted into a discrete digital value (usually 0-255 for an 8-bit ADC).

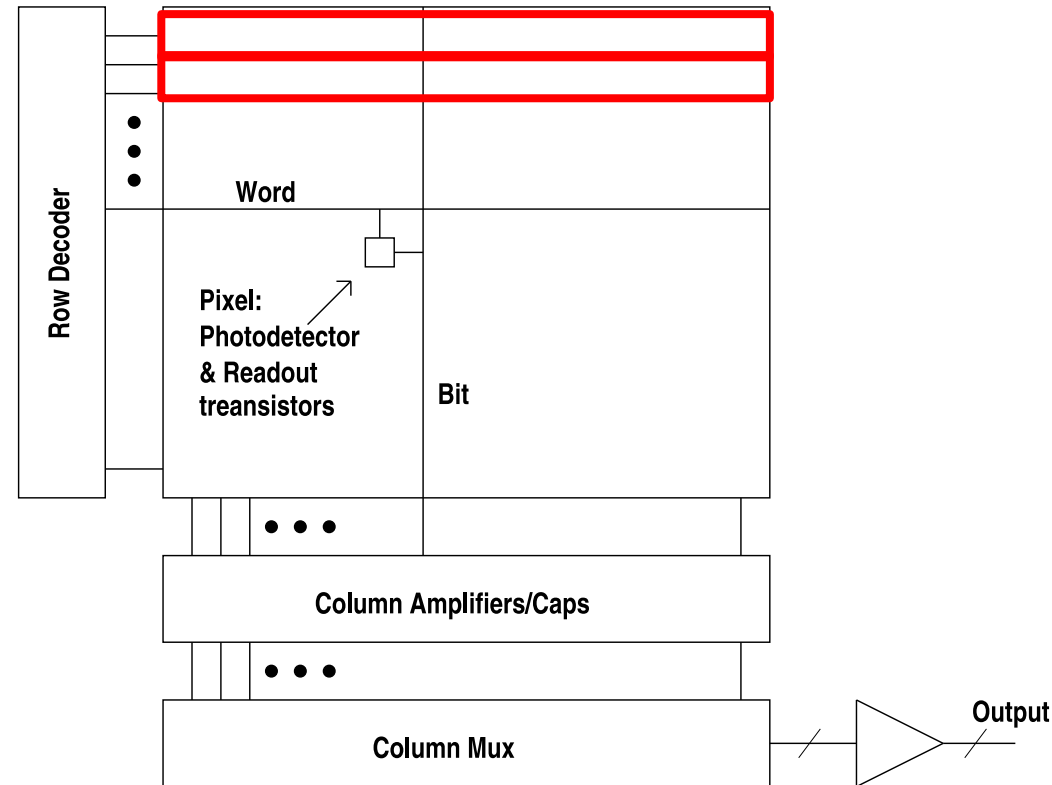
CMOS Image Sensor Integrated Circuit Architecture



<https://evidentscientific.com/en/microscope-resource/knowledge-hub/digital-imaging/cmosimagesensors>

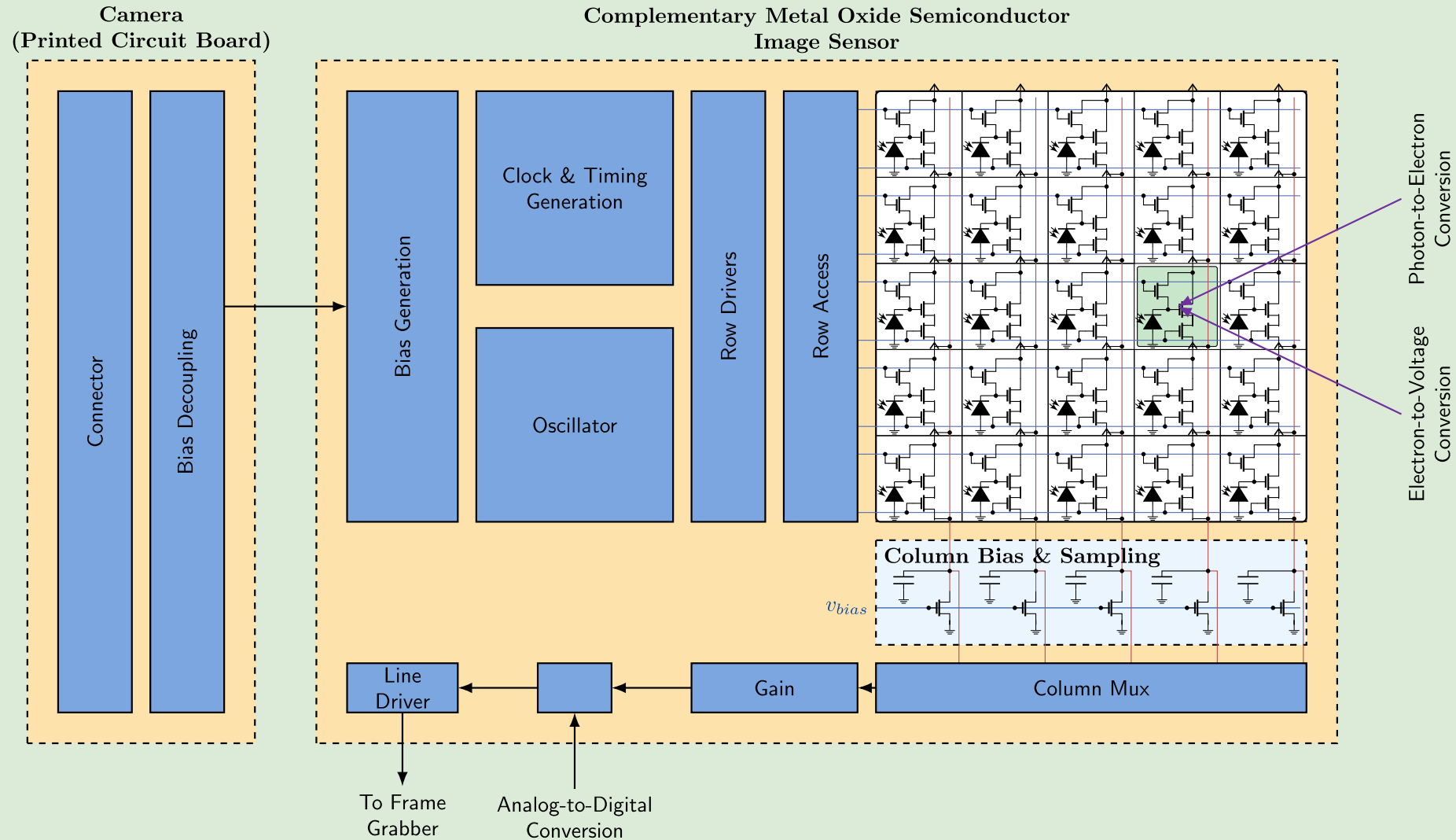


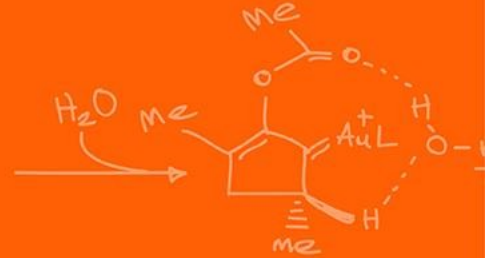
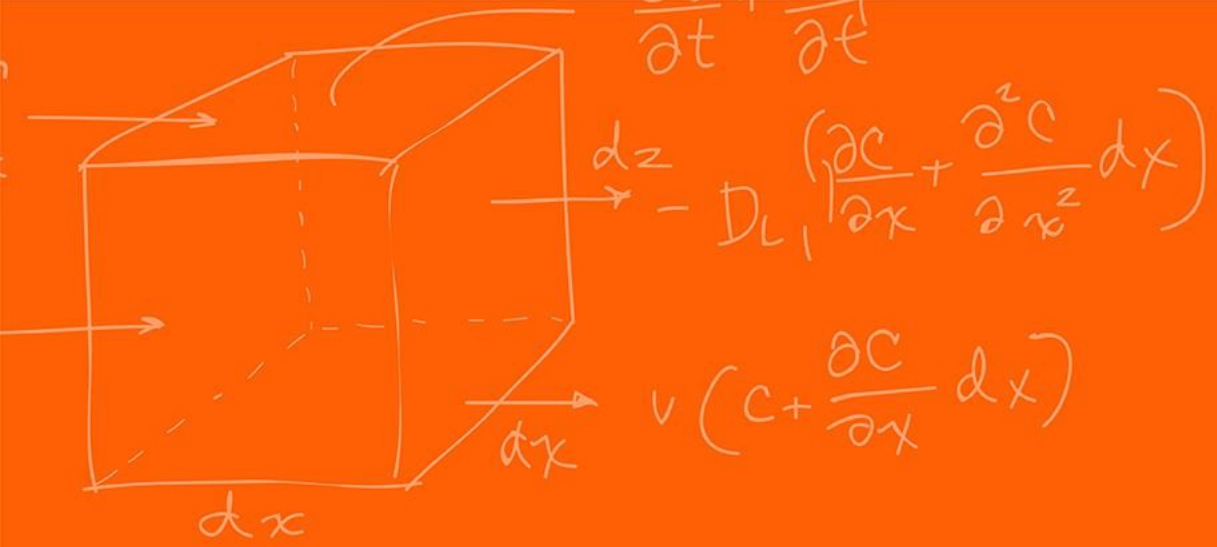
<https://evidentscientific.com/en/microscope-resource/knowledge-hub/digital-imaging/cmosimagesensors>



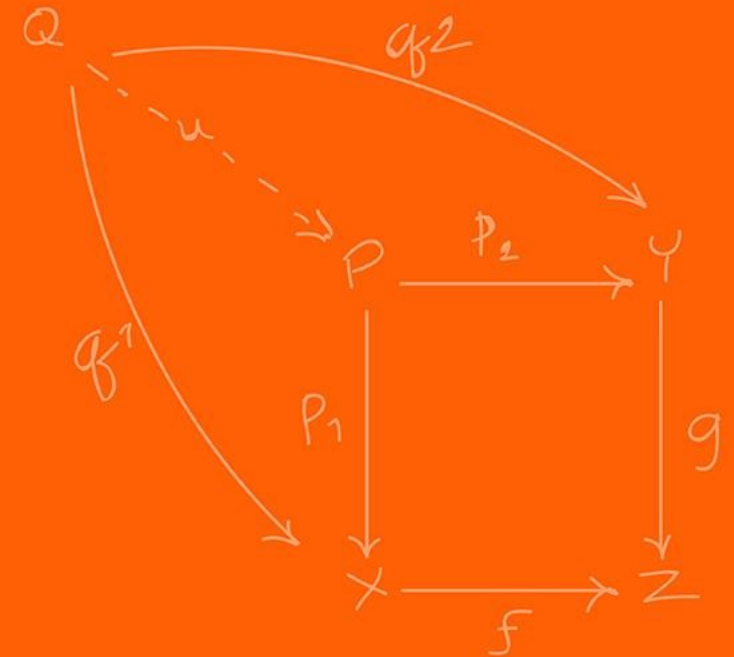
<https://isl.stanford.edu/~abbas/ee392b/lect04.pdf>

- Readout performed by transferring one row at a time to the column storage capacitors, then reading out the row, one (or more) pixel at a time, using the column decoder and multiplexer
- In many CMOS image sensor architectures, row integration times are *staggered* by the row / column readout time (scrolling shutter)





Color



The problem with photodiodes is that they only measure the intensity of light (how many photons hit them).

What kind of problem that cause?

They are inherently colorblind; they only produce a grayscale image!

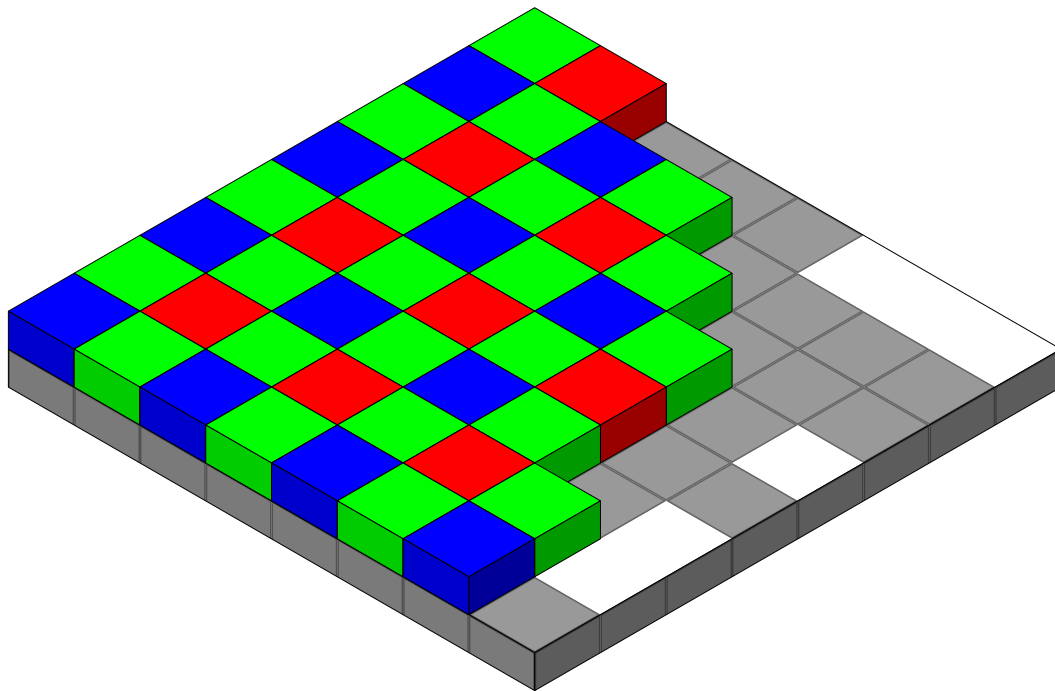
To see a red ball, we need a way to filter out all other wavelengths of light before they hit the photodiode.

How do we make a single sensor array see Red, Green, and Blue simultaneously?

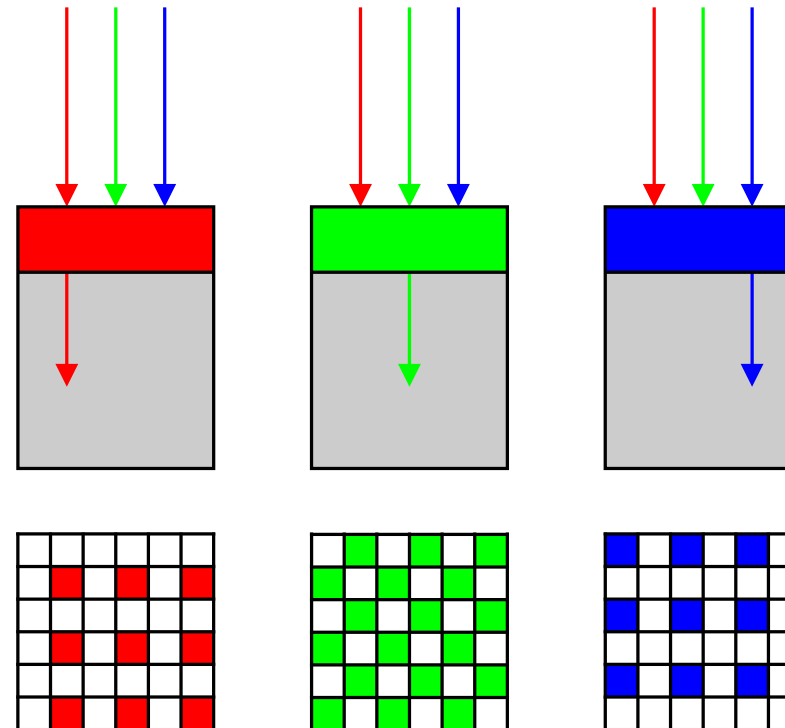
To achieve color vision, a microscopic color filter is placed over each individual photodiode.

This is known as a Color Filter Array (CFA), the most common being the Bayer Filter.

It assigns each pixel to capture only Red, only Green, or only Blue light.



https://en.wikipedia.org/wiki/Bayer_filter



The Bayer format arranges filters in a repeating 2×2 grid.

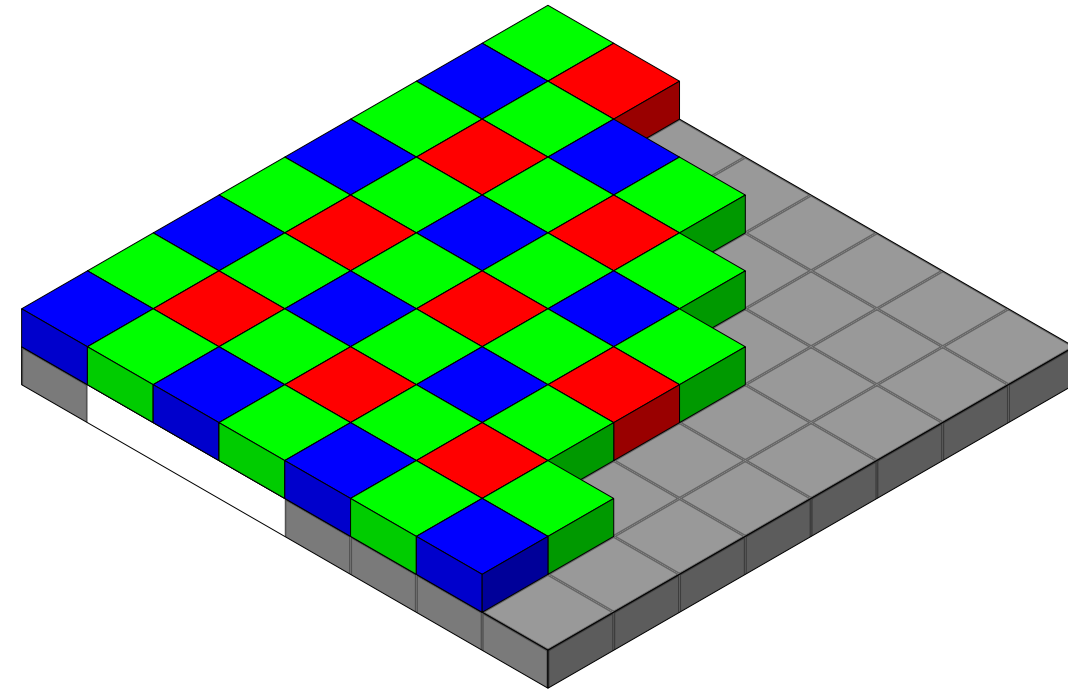
What is a problem with filtering color?

We lose information! This is why most scientific instruments (on Mars for example) are in black and white to ensure capturing all the intensities!

Notice anything strange about this ratio?

1 Red : 2 Green : 1 Blue

Why are there twice as many Green pixels as Red or Blue?



https://en.wikipedia.org/wiki/Bayer_filter

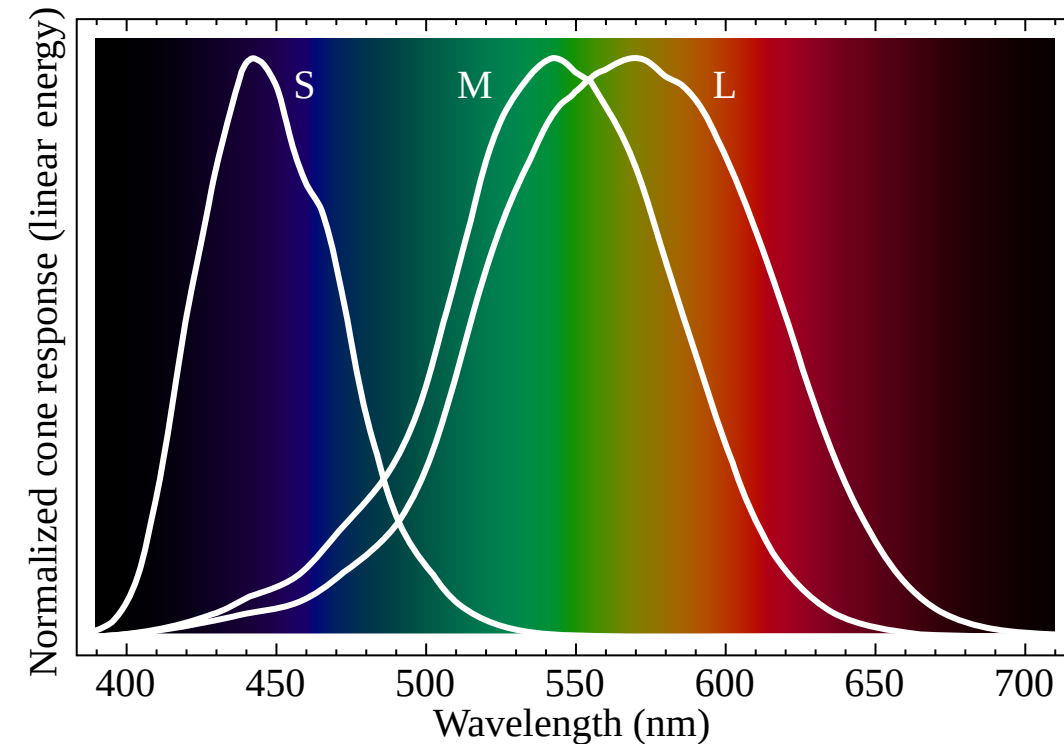
We use twice as many green filters because the human eye (specifically our cone cells) is most sensitive to green wavelengths (~550 nm).

By over-sampling green, the camera captures luminance (brightness) detail that matches human perception.

However, our raw array is now a mosaic of single colors.

Pixel (0,0) only knows Red.

How do we get RGB values for every pixel?



https://en.wikipedia.org/wiki/Color_vision

Demosaicing is the mathematical process of guessing the missing color values at each pixel location.

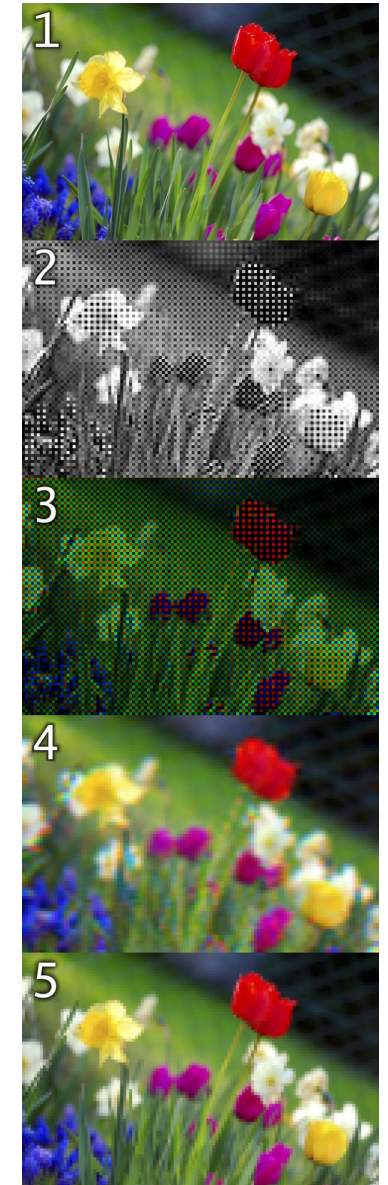
We use surrounding pixels to calculate an average. To find the missing Green value at a Red pixel:

$$R_{i,j}: G_{i,j} = \frac{1}{4} (G_{i-1,j} + G_{i+1,j} + G_{i,j-1} + G_{i,j+1})$$

The camera's internal DSP (Digital Signal Processor) usually does this before sending data to the microcontroller.

This simple approach works well in areas with constant color or smooth gradients, but it can cause artifacts such as color bleeding in areas where there are abrupt changes in color or brightness especially noticeable along sharp edges in the image.

Because of this, other demosaicing methods attempt to identify high-contrast edges and only interpolate along these edges, but not across them.



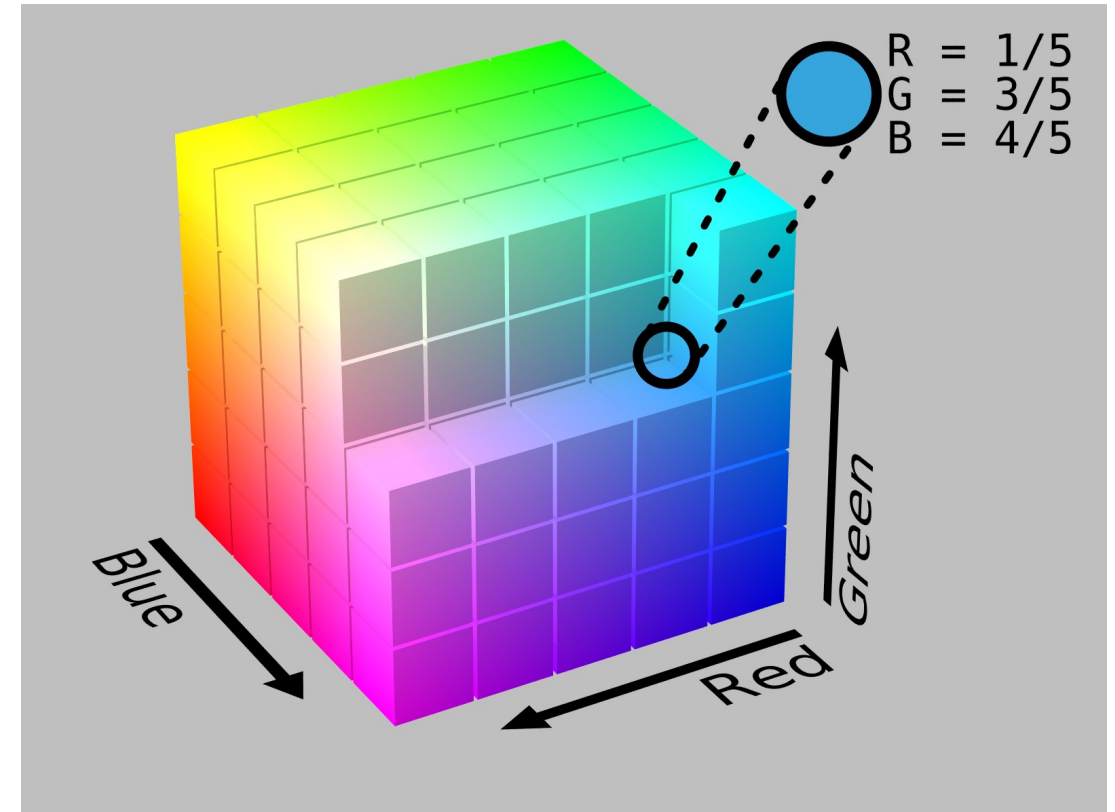
Once demosaiced, every pixel is an array of three values: **R**, **G**, **B**.

RGB is an additive color model used by screens and sensors and each value is represented as an 8-bit value (depends on the ADC of the sensor).

If $R = 255, G = 255, B = 255$, what color do we see?
White

If $R = 255, G = 0, B = 0$, what color do we see?
Red

RGB is represented as a 3D Cartesian cube, where x, y, z correspond to Red, Green, and Blue intensities.



Imagine we write code to track a red ball:

```
if (R > 200 && G < 50 && B < 50) { track(); }
```

What happens when the robot drives under a shadow?

The total light drops. The pixel might become R=100, G=10, B=10. Our code fails!

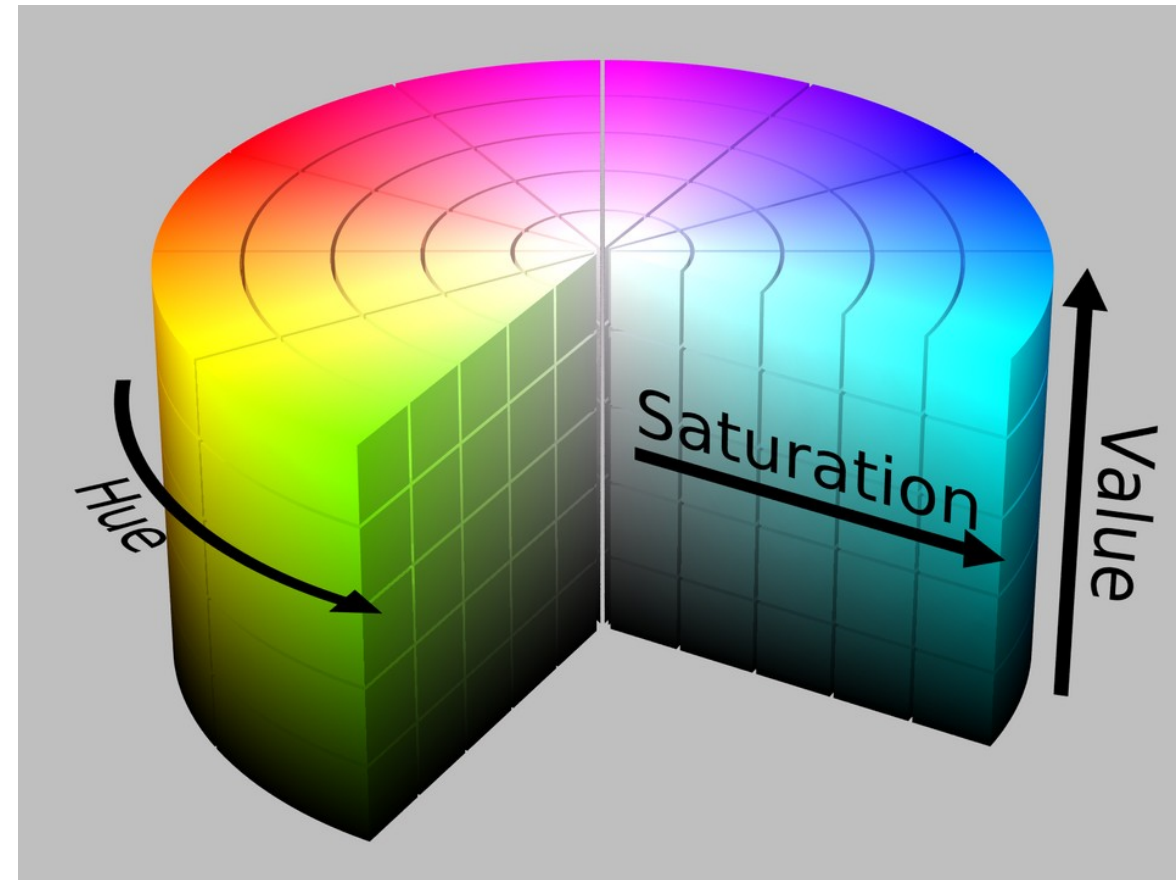
In RGB, a change in lighting changes all three variables simultaneously. It couples chrominance (color) with luminance (brightness).

To solve the shadow problem, we transform the RGB cube into a cylinder: The **HSV** color space.

H = **Hue** (The actual color type, e.g., Red, Blue).

S = **Saturation** (How pure the color is, vs. faded white).

V = **Value** (How bright or dark the color is).



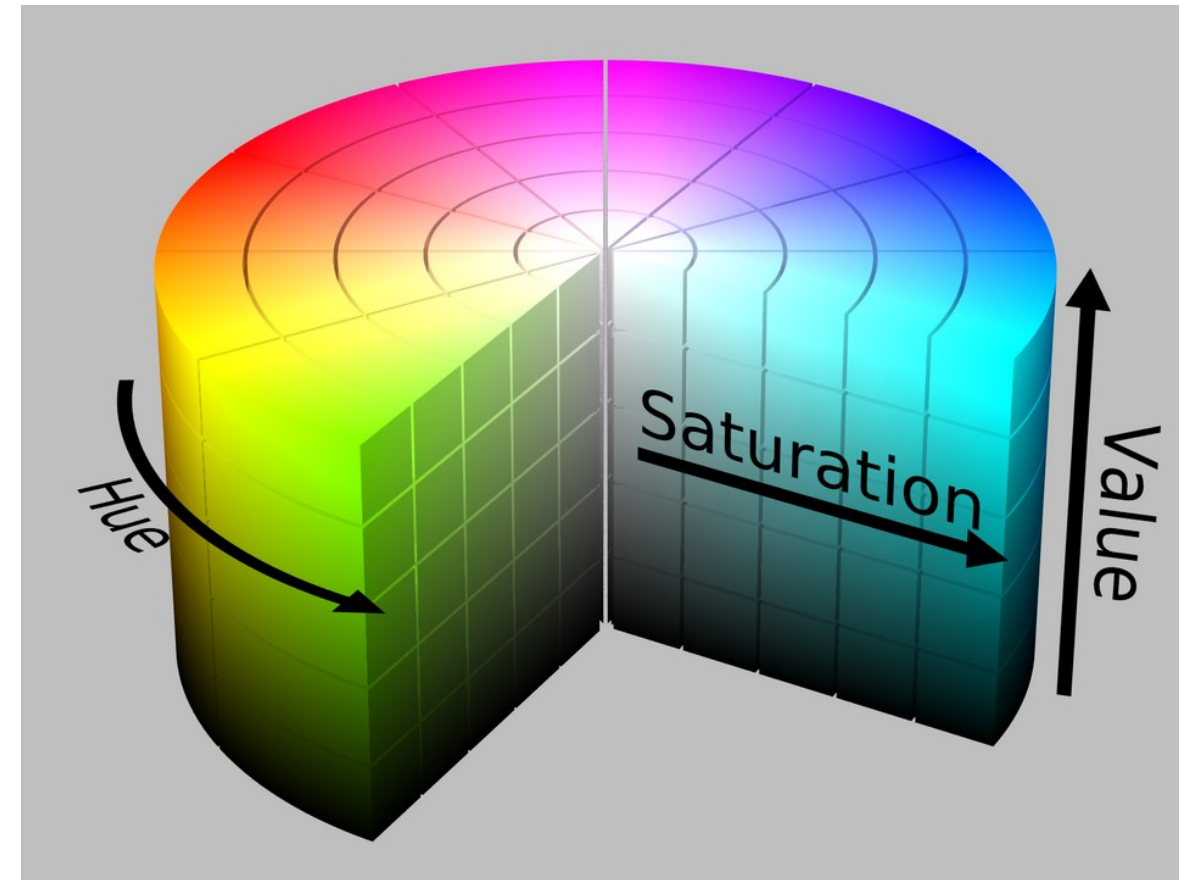
https://en.wikipedia.org/wiki/HSL_and_HSV

• In the HSV cylinder, Hue is an angle from 0° to 360° .

- 0° = Red,
- 120° = Green,
- 240° = Blue.

Why this is brilliant: If a red ball enters a shadow, its **Value drops**, but its **Hue angle remains “exactly” 0° !**

This isolates the color from the lighting condition, making our tracking algorithm more robust!



https://en.wikipedia.org/wiki/HSL_and_HSV

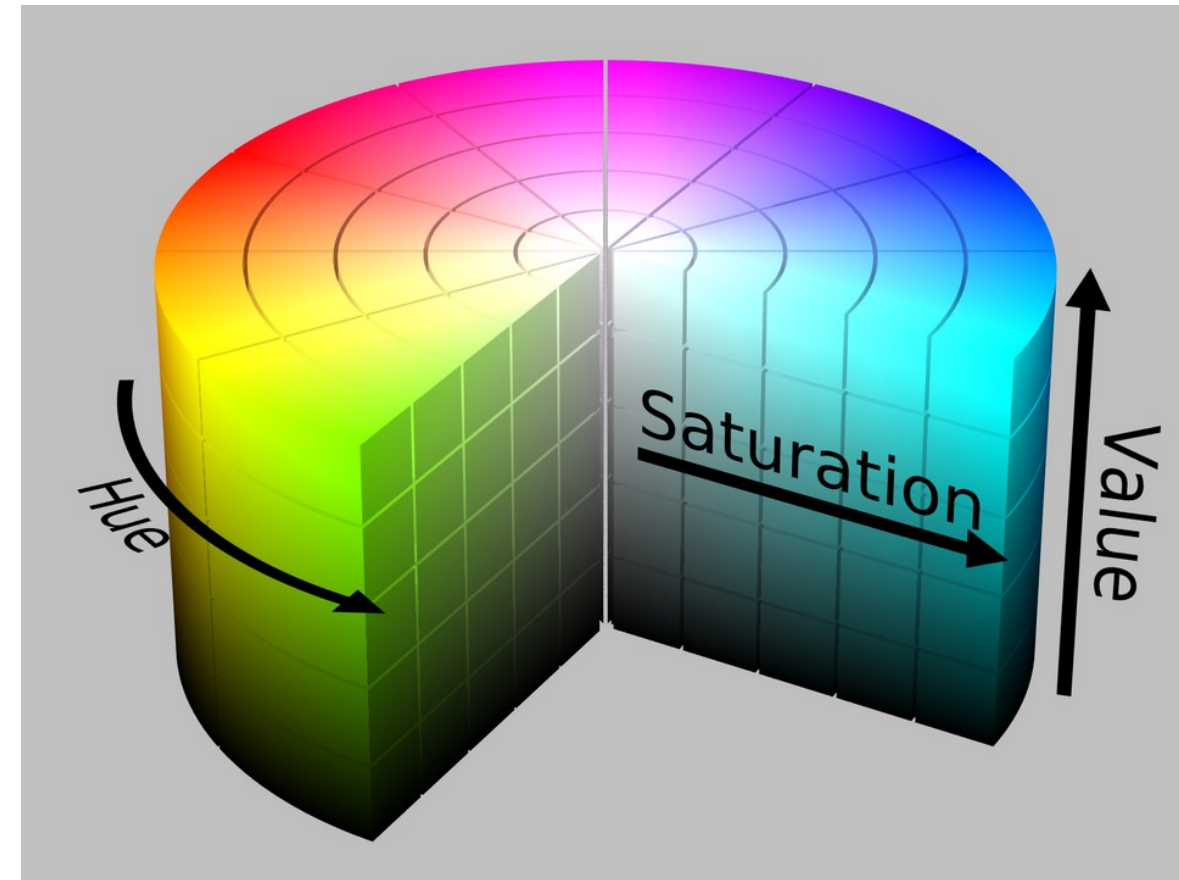
Saturation is the radius (distance from the center axis).

- **S=0%** is pure gray/white (center).
- **S=100%** is a pure, vibrant color (edge).

Value is the height of the cylinder (0 to 100%).

- **V=0%** is pitch black.
- **V=100%** is fully illuminated.

To track an object, we look for pixels that fall within specific ranges.



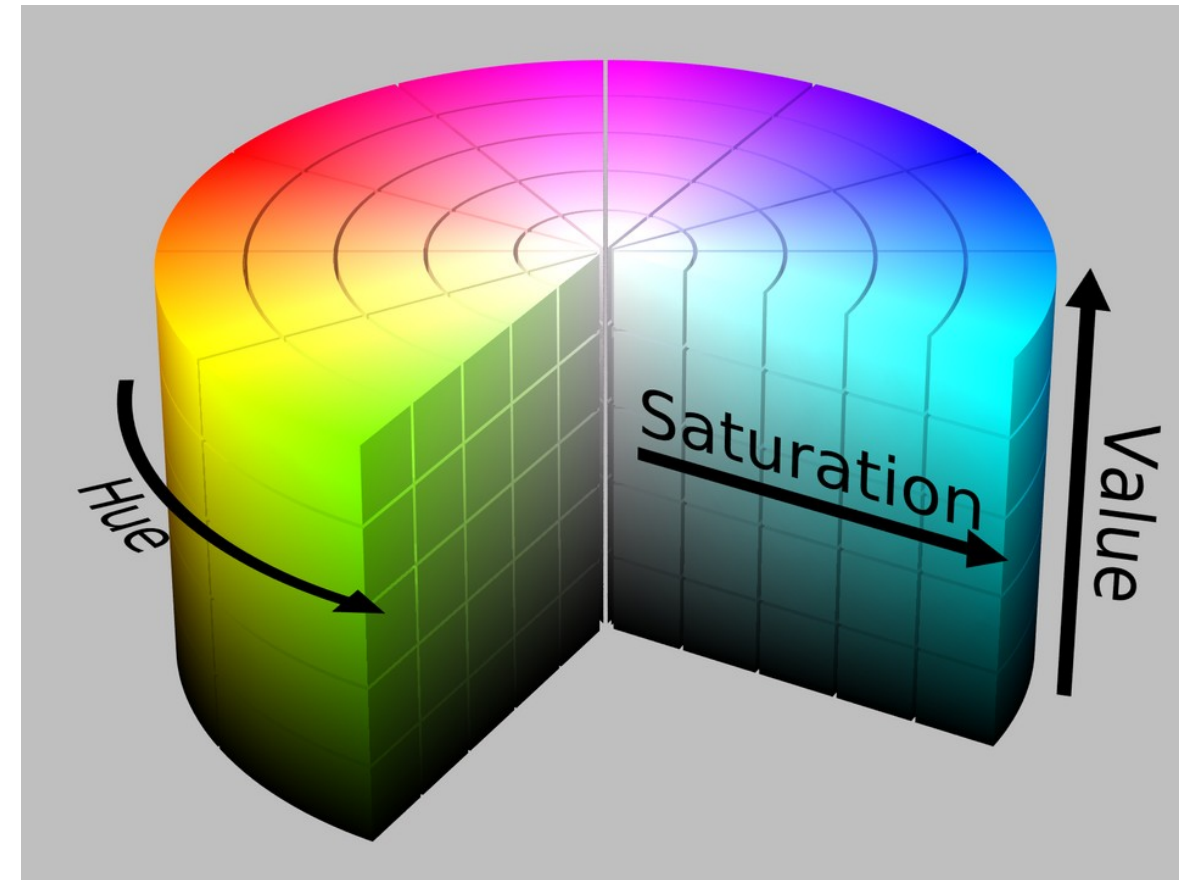
https://en.wikipedia.org/wiki/HSL_and_HSV

To convert from RGB to HSV the conversion is simple.

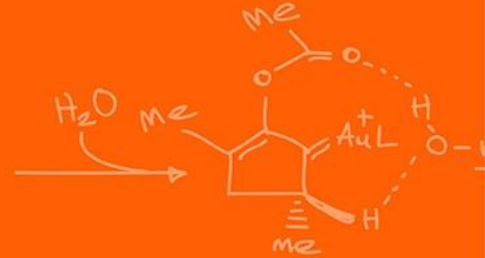
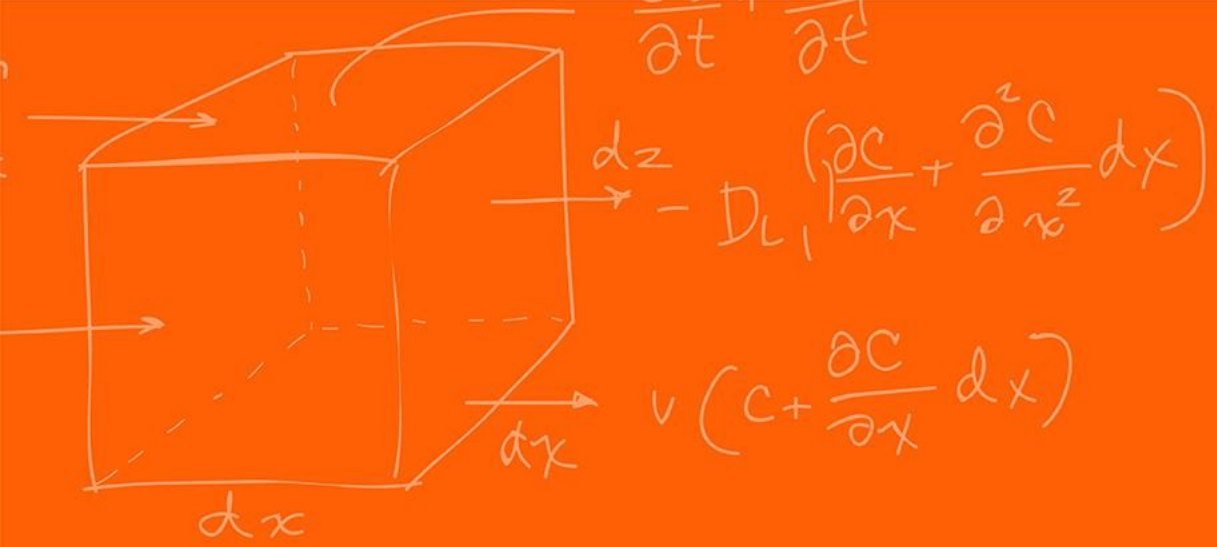
Let R', G', B' be normalized values in $[0,1]$.

$$C_{max} = \max(R', G', B'), \quad C_{min} = \min(R', G', B'), \\ \Delta = C_{max} - C_{min},$$

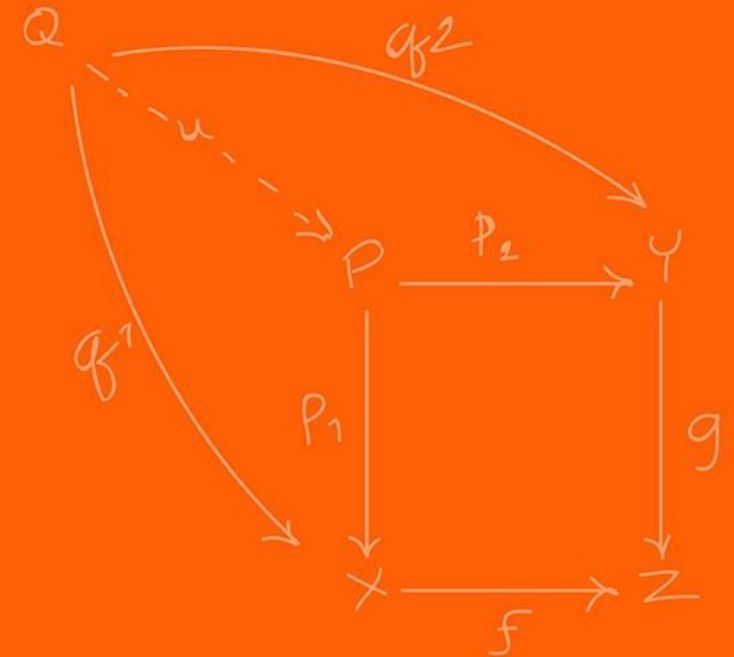
$$V = C_{max}, \\ S = \begin{cases} 0, & C_{max} = 0 \\ \frac{\Delta}{C_{max}} & \end{cases}, \\ H = 60^\circ \times \begin{cases} \frac{G - B}{\Delta} \% 6, & C_{max} = R \\ \frac{B - R}{\Delta} + 2, & C_{max} = G \\ \frac{R - G}{\Delta} + 4, & C_{max} = B \end{cases}$$



https://en.wikipedia.org/wiki/HSL_and_HSV



Color



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IMAGE / GRAPHIC

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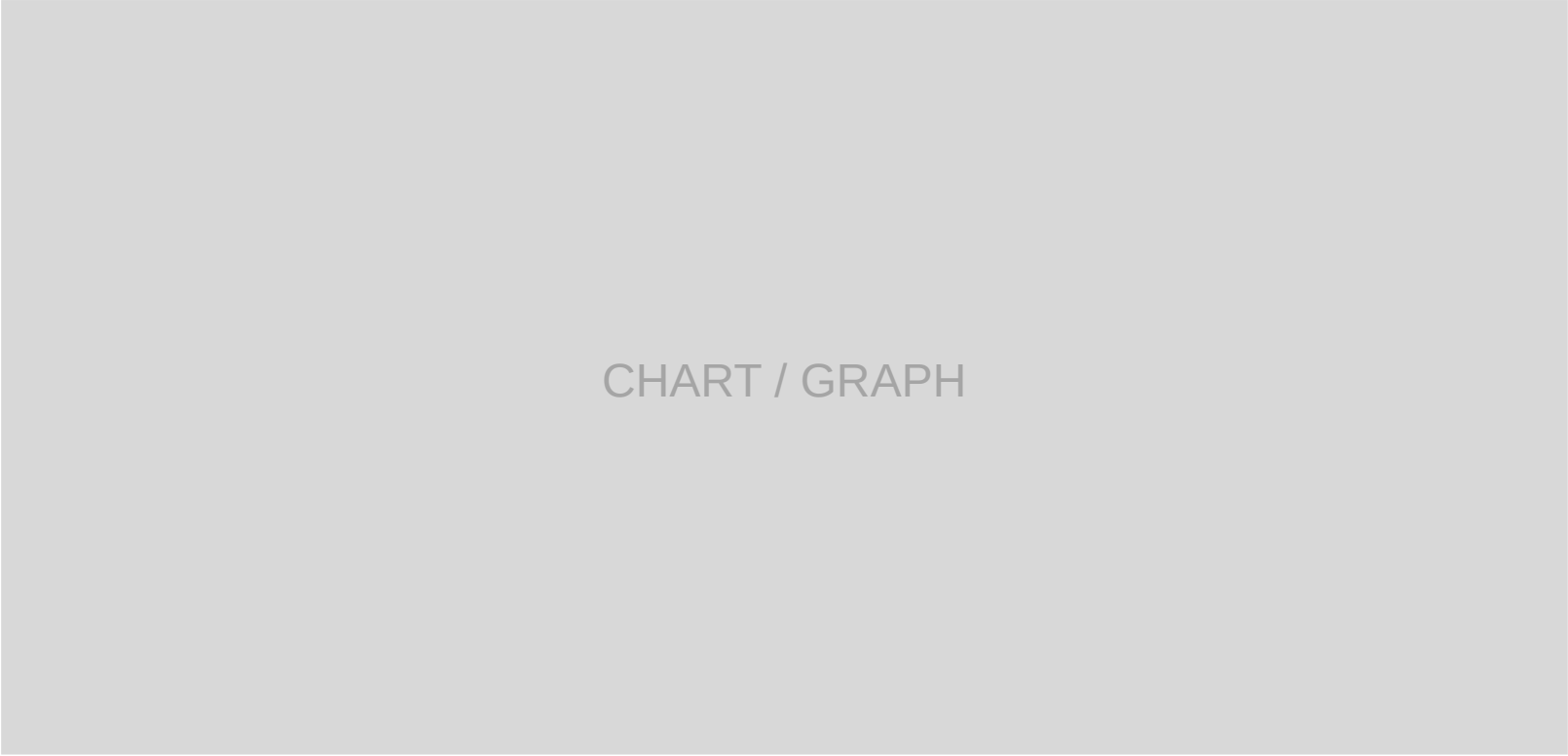
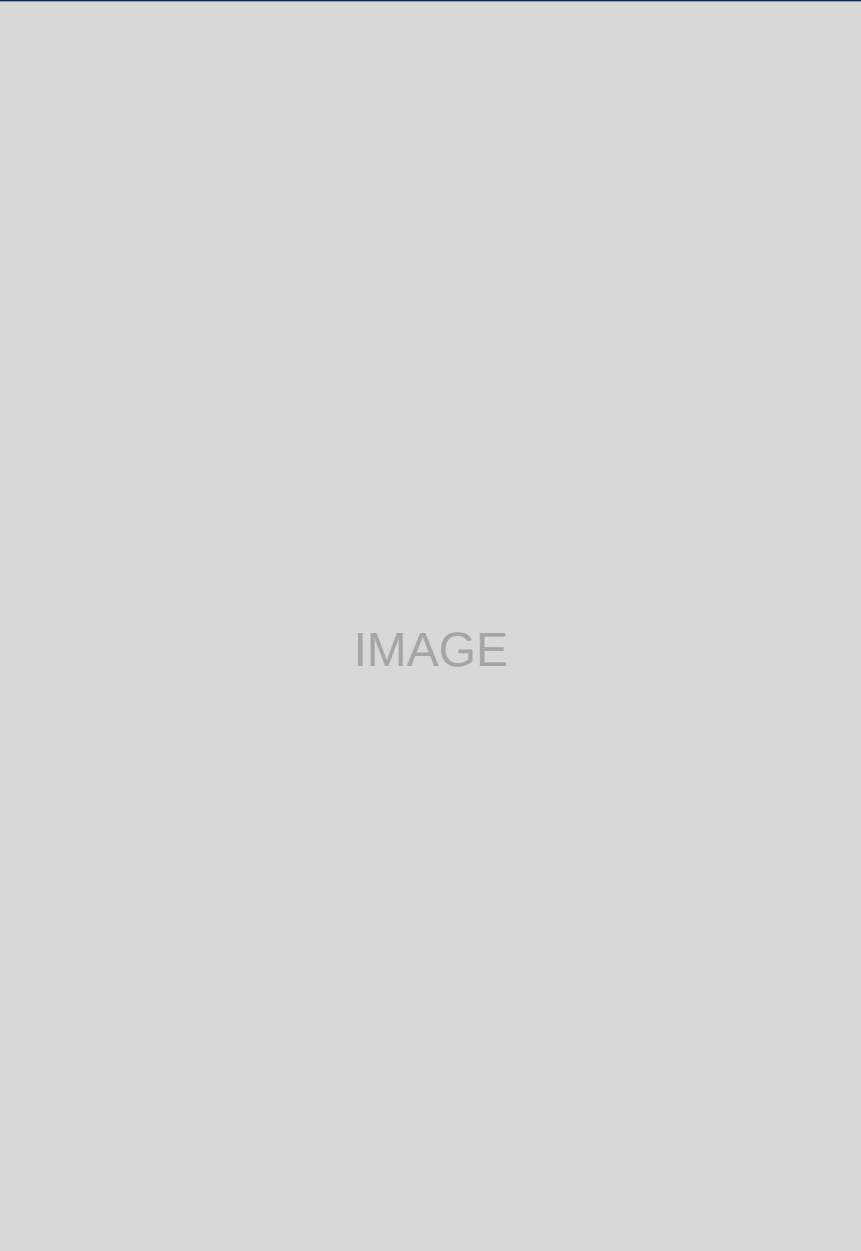


CHART / GRAPH

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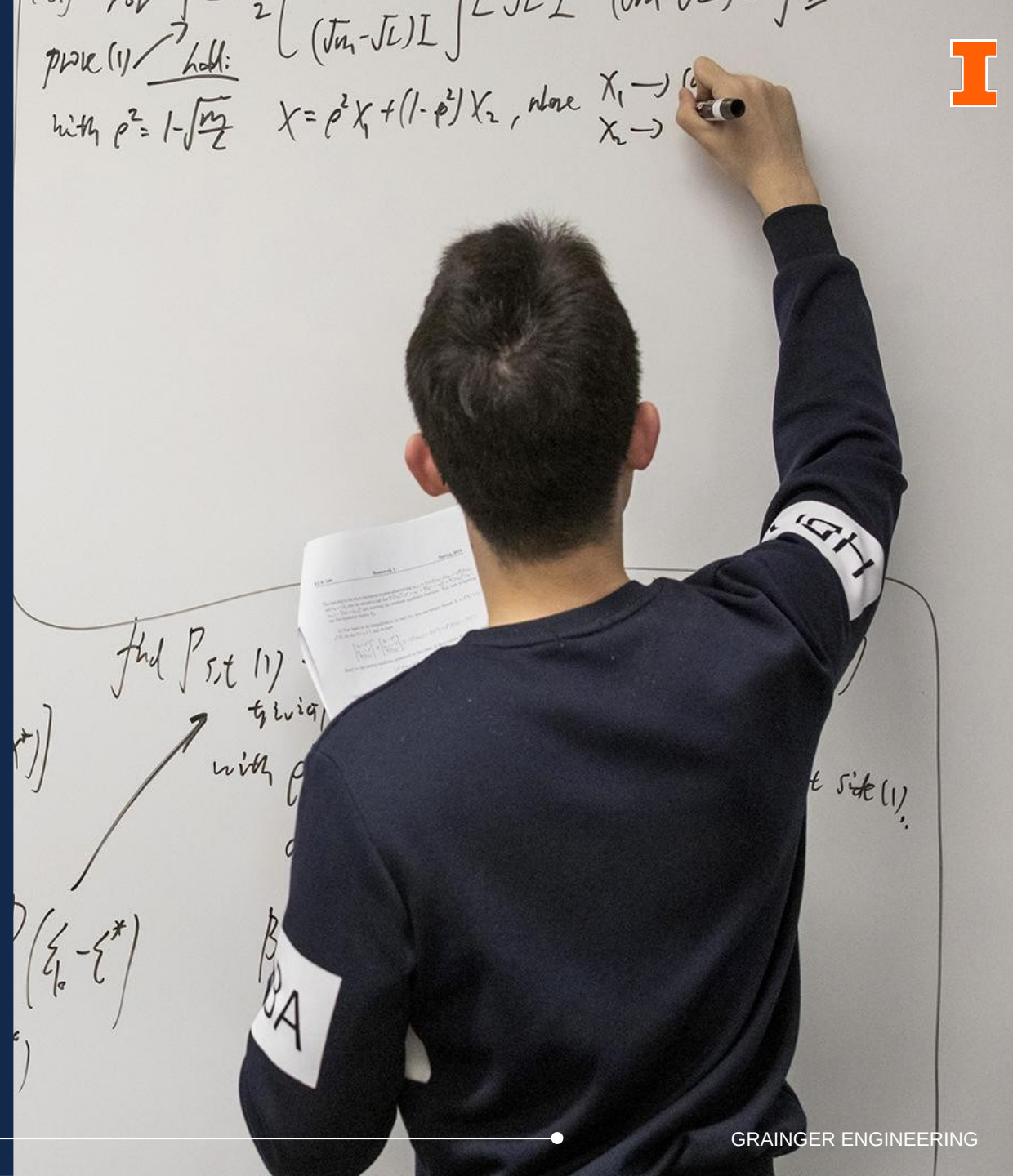
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IMAGE

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(Closing slide option. Suggestions for last slide)

Summary

Thank You

Questions

Contact Information

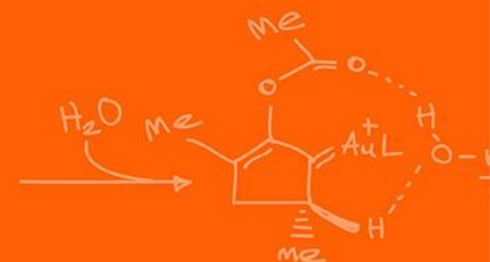
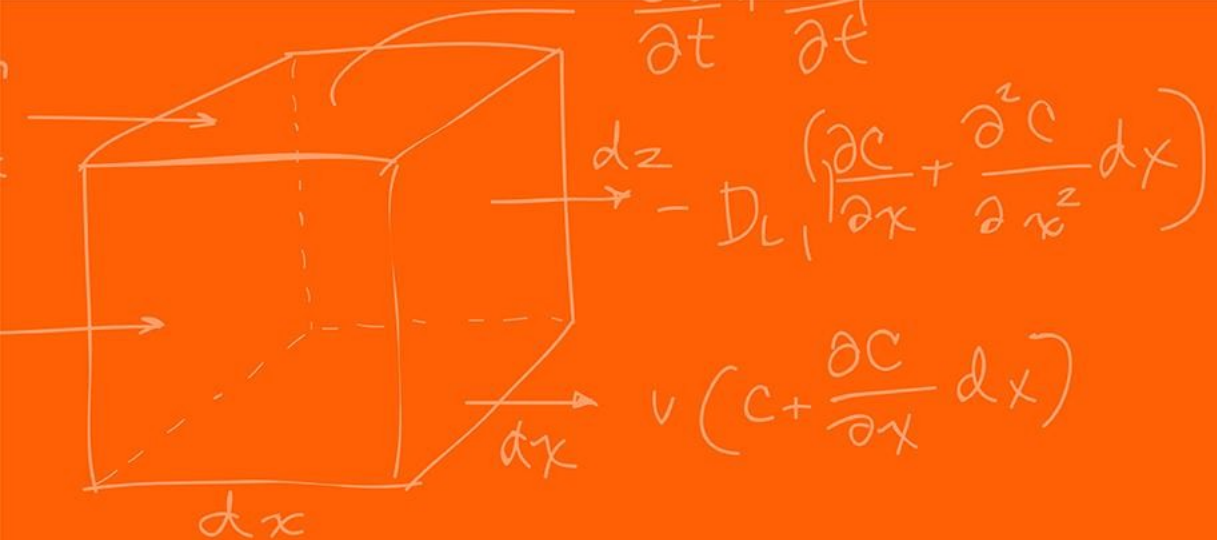
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Questions?



The Grainger College of Engineering

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